5 (a) (i) $2\pi f = 377$ [M1]

$f = 60$ Hz [A1]

(ii) $V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{240}{\sqrt{2}}$ [M1]

$= 170$ V [A1]

(iii) Alternating magnetic field is created in the coil.
Hence there is an <u>alternating magnetic flux linkage </u>through the ring [B1]

By Faraday's law<u>, an e.m.f. will be induced in the ring</u> [B1 – link to emf to flux linkage]

Since the ring is a <u>closed loop</u>, <u>current is induced</u> in the ring. [B1]

By Lenz's law, the <u>current will flow in a direction to oppose the changes in the magnetic flux linkage</u>, hence creating a (average opposing) <u>force which balances the weight</u>. [B1]

(iv) Since the ring is <u>not a closed loop</u>, <u>current is not induced</u> in the ring. [M1]

Hence no opposing force is created, ring <u>does not float.</u> [A1]

(b)

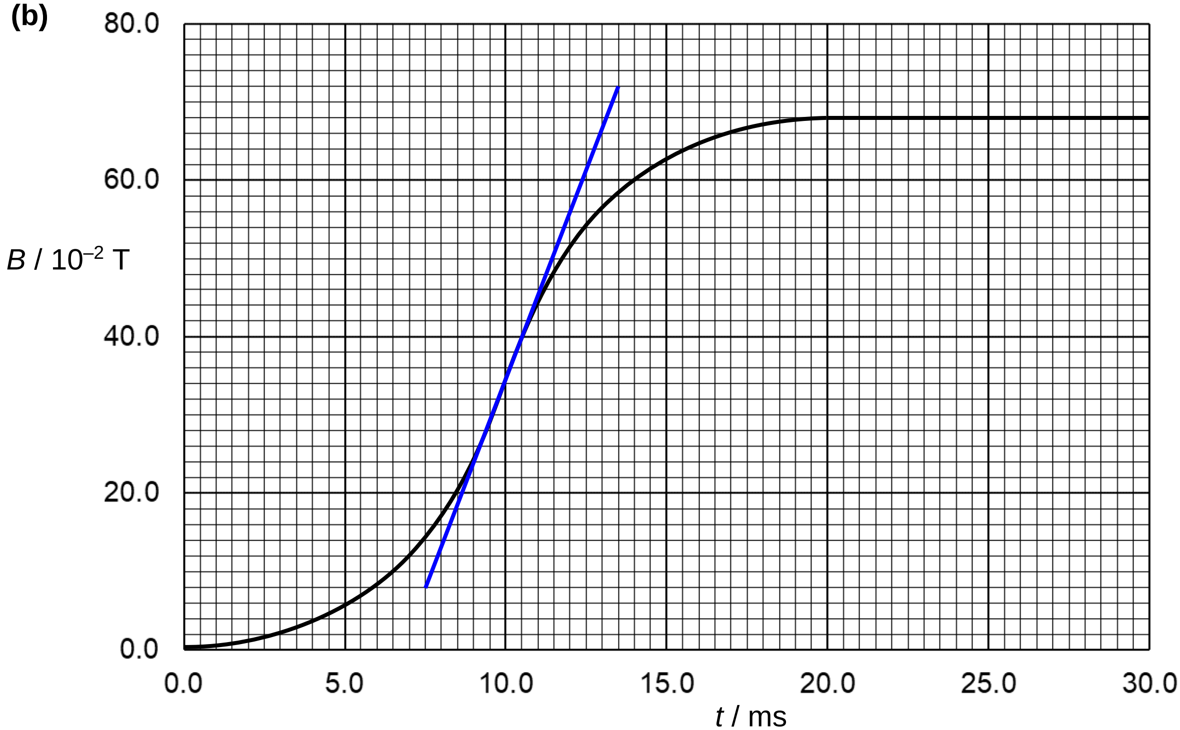

Fig. 5.4

Maximum e.m.f. will be induced when the gradient is steepest, i.e. at 10 ms.

$$Gradient = \frac{dB}{dt} = \frac{(72.0 - 8.0)\times 10^{-2}}{(13.5 - 7.5)\times 10^{-3}} = 106.67 \quad \text{[M1]}$$

$$\left|\varepsilon_{\text{max}}\right| = \frac{d(N\phi)}{dt} = NA\frac{dB}{dt} = (80)\left(4.2\times 10^{-5}\right)(106.67) = 0.3584 = 0.36\text{ V} \quad \text{[A1]}$$

Allow variation in value of gradient from 100 to 112 (which relates to a final answer for maximum e.m.f to be from 0.336 to 0.376 V)
–1 mark if gradient triangle is less than one big square (vertically)